

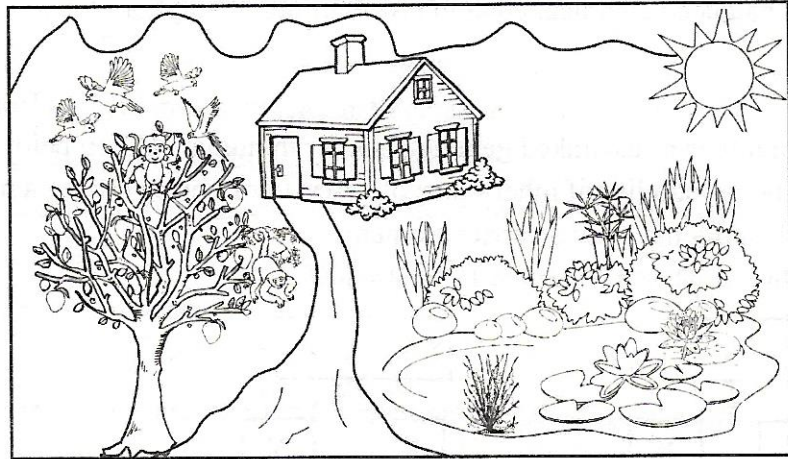
බස්නාහිර පළාත් අධ්‍යාපන දෙපාර්තමේන්තුව மேல் மாகாணக் கல்வித் திணைக்களம் Department of Education - Western Province			
වර්ෂ අවසාන ඇගයීම ஆண்டிறுதி மதிப்பீடு - 2018 Year End Evaluation			
ශ්‍රේණිය } 10 தரம் } Grade	විෂයය } Science பாடம் } Subject	පත්‍රය } II வினாத்தாள் } Paper	කාලය } 03 hours காலம் } Time
Name :-		Index No. :-	

Instructions:-

- ♦ Answer all the question in part A in the space provided.
- ♦ Answer 3 questions in part B.

Part A - Structured Essay Type

(01) (A) There is a fresh water pond and a large mango tree in Thamali's home garden. Water in the pond has been turned in to green. There are blue lotus and Valisnaria plants among the aquatic plants in the pond. There were monkeys and birds on the mango tree.



- (i) Write a common living characteristic of the mango tree and the monkey. (M. 01)
 - (ii) Write a factor plant obtain from external environment for the process of production of food. (M. 01)
 - (iii) Mention a group of micro organism that can be observed, if green coloured water sample was viewed through a microscope.(M. 01)
 - (iv) When classifying animals according to the mode of nutrition, What is the mode of nutrition monkeys show? (M. 01)
 - (v) How would the pollination of valisnaria flowers take place?(M. 01)
- (B) Nimal lighted a (fire skier) sky rocket cracker from the mango tree. Not only monkeys but all the other animals, even birds left the tree.
- (i) Name the living characteristic which is relevant to the instance of monkeys jumping away from the tree?..... (M. 01)

- (ii) Write one method of co - ordination between different organs when responding to a stimulus..... (M. 01)

(C) Given here is a sketch diagram of a fire skier (sky rocket cracker)

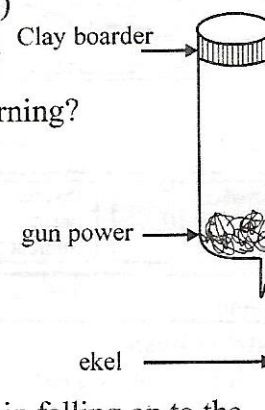
- (i) What is the Newton's law that describe the motion of the sky rocket cracker when it is in motion after gun power starts burning?

.....(M. 01)

- (ii) Write an example for another instance where above mentioned law is used(M. 01)

- (iii) State 2 forces which act on the sky rocket cracker to move it upwards (M. 02)

- (iv) What is the nature of motion the sky rocket cracker undergo in falling on to the earth after burning?.....(M. 01)



(D) Sulphur is a component of gun powder.

- (i) Write 2 observations when burning a piece of sulphur.

..... (M. 02)

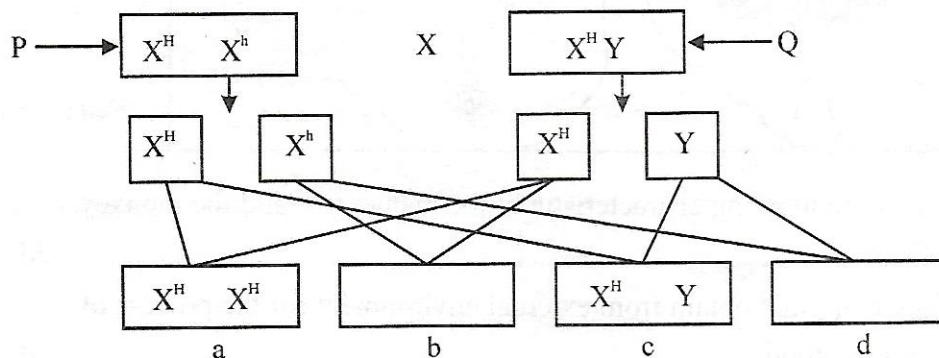
- (ii) Write the balanced chemical equation for burning of sulphur

..... (M. 01)

(Total marks - 15)

- (02) (A) A Haemophila is a sex-linked genetic disorder in human. Given below is a sketch to show the probability of inheritance of genes to offspring during a marriage between healthy man and a carrier women.

('h' recessive condition is responsible for the disease.)



- (i) Out of P and Q what is the letter representing the man?.....(M. 01)

- (ii) Write the genotypes for b and d

b - d - (M. 02)

- (iii) According to the above diagram what is the probability of having a healthy female child? (M. 01)

- (iv) Write the main symptom of haemophilia.

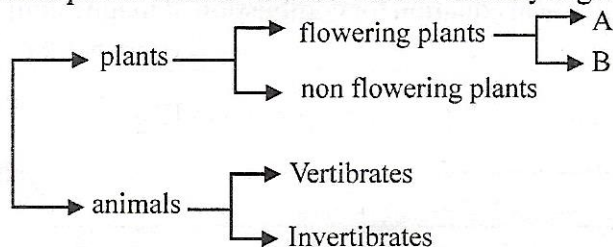
..... (M. 01)

- (v) There is a belief that there is a risk of marriages between blood relatives. State the reason for this belief (M. 01)
- (vi) What kind of a hereditary disease is Albinism? (M. 01)

(B) Many methods are used to classify organisms.

- (i) Who put forward the 3 domain classification? (M. 01)
- (ii) What is the domain to which eukaryotes belong? (M. 01)
- (iii) State one main characteristic of natural classification.
..... (M. 01)
- (iv) Write a use of scientific classification of organism
..... (M. 01)

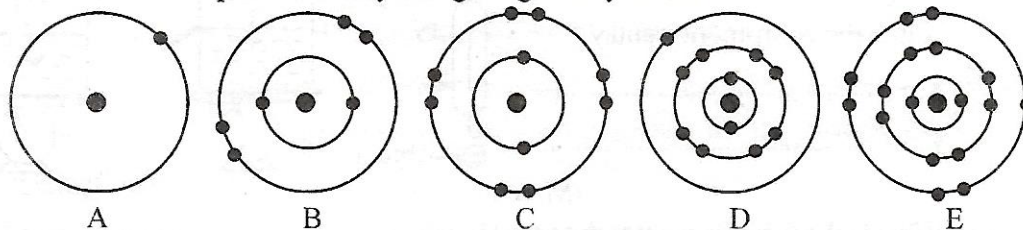
(C) Given below is a part of a chart which is used to classify organisms.



- (i) Name two plant groups which can be categorized as A and B
A- B- (M. 01)
- (ii) Write 2 common characteristics which can only be seen in vertebrates
1- 2- (M. 02)
- (iii) Name a non flowering plant with naked seeds (M. 01)

(Total marks - 15)

(03) (A) Given below are structures of few elements among first 20 elements in the periodic table. Answer the questions only using the given symbols

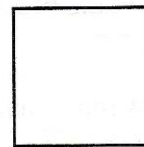


- (i) Who is the scientist present the nuclear model of atom?
..... (M. 01)
- (ii) Name the pair of elements which belong to 3rd period (M. 01)
- (iii) What is the letter corresponding to the element which denotes an allotropic form which conducts electricity (M. 01)
- (iv) Write a physical property of element D (M. 01)

(B) What is the type of bond present in the compound formed by combining elements D and E?

..... (M. 01)

(ii) Draw the Lewis structure of the molecule formed by combining A and B



(M. 01)

(iii) Explain what are isotopes

..... (M. 01)

(C) 3 g of cleaned Magnesium was heated strongly in a crucible until it burned

(Mg = 24) (O = 16)

(i) Write and observation when burning magnesium

..... (M. 01)

(ii) Write the balanced chemical equation for combustion of magnesium

..... (M. 02)

(iii) What is the number of moles of Magnesium present in 12 g?

.....
..... (M. 02)

(iv) What is the mass of oxygen combining with 12 g of magnesium?

.....
..... (M. 02)

(v) What is the number of atoms in 12 g of magnesium?

..... (M. 01)

(Total marks - 15)

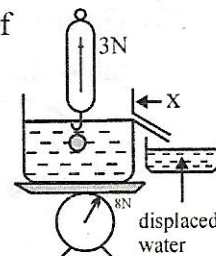
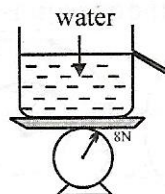
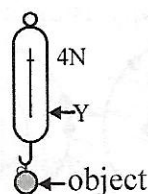
(04) (A) The following is an apparatus arranged to investigate the validity of a certain law

(i) Name the equipment x and y

x -

y -

(M. 02)



(ii) What is the law that is experimented here? (M. 01)

(iii) Write relationship stated by the law mentioned in (ii) above

..... (M. 01)

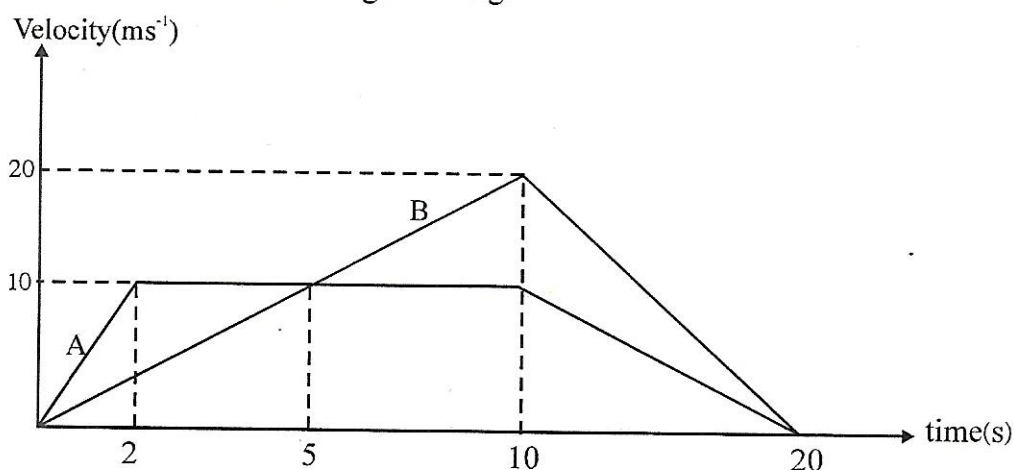
(iv) What is the upthrust exerted on the object?

..... (M. 01)

(v) Calculate the mass of displaced volume of water, ($g = 10ms^{-2}$)

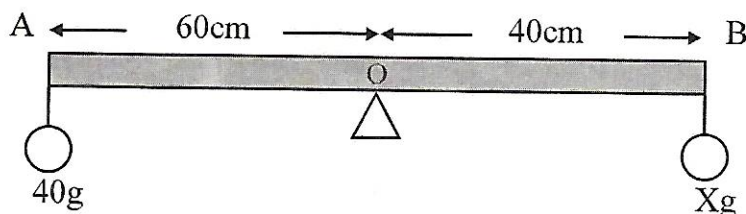
.....
..... (M. 02)

- (B) The following is a Velocity-time graph showing the motion of two competitors A and B who were running in a straight line



- How many seconds after starting the game the competitors show an equal velocity?
..... (M. 01)
- Calculate the acceleration of player B within 10 s
.....
..... (M. 02)
- Find the distance travelled by A with a uniform velocity
..... (M. 01)

- (C) Diagram shows how a uniform rod is balanced at point 'O'



- When the rod is balanced at 'O', calculate the force exerted by x
.....
..... (M. 02)
- The rod should be balanced at 'o' after hanging a mass of 60 g instead of 40 g at the end A. What is the distance 60 g mass needs to be moved from A towards O?
.....
..... (M. 02)

(Total marks - 15)

Part B - Essay Type

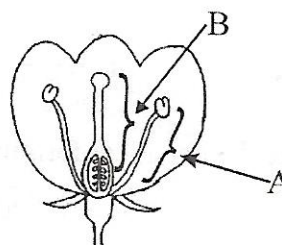
Instructions :- Answer 3 questions only

- (05) (A) The table shows the details of food tests carried out to investigate on presence of organic molecules in living matter, The table is incomplete

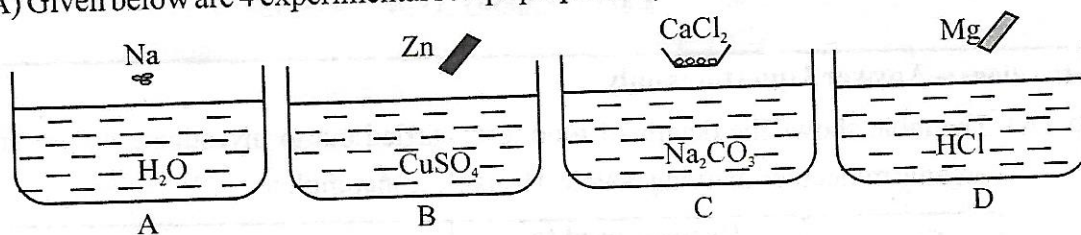
Food	Reagent used to investigate	Observation
A	Bendict solution	After sevaral colour changes, it takes brick colour at the end
B	Sodium hydroxide and copper sulphate solution (Biurette Solution)	X
C	Y	Red colour globules floats over the surface of the solution

Answer the following question using the information given above

- (i) What is the other step which should be carried out to get the above mentioned colour change after adding benedicts solution into food item A (M. 01)
 - (ii) Write the colour change which takes place in A in respective order (M. 01)
 - (iii) Write an example for a suitable food item to which B can be used (M. 01)
 - (iv) Mention the observation at X and reagent used at Y (M. 02)
 - (v) Write 2 significance of biomolecules done by present in C (M. 02)
 - (vi) Write one useful function carried out by bio molecule DNA (M. 01)
- (B) Given below is a generalized diagram of a flower which shows 2 major parts
- (i) Give common names for A and B (M. 02)
 - (ii) Write the part of a flower which contribute in following functions (M. 02)
 - a - attracting insects
 - b - protecting immature floral parts
 - (iii) What is meant by pollination? (M. 01)
 - (iv) Mention the process that takes place at the stigma after pollination (M. 01)
 - (v) Write 2 differences that takes place in a flower after fertilization of ova (M. 01)
 - (vi) Write the importance of dispersed of seeds and fruits. (M. 01)
- (C) There are two methods of reproduction present in organism. They are sexual reproduction and Asexual reproduction.
- (i) Write 2 differences by which asexual reproduction differ from sexual reproduction (M. 02)
 - (ii) What is the place where male gametes are produced in male reproductive system? (M. 01)
 - (iii) Name the hormone which produces in reproductive system of males (M. 01)
- (Total marks - 20)

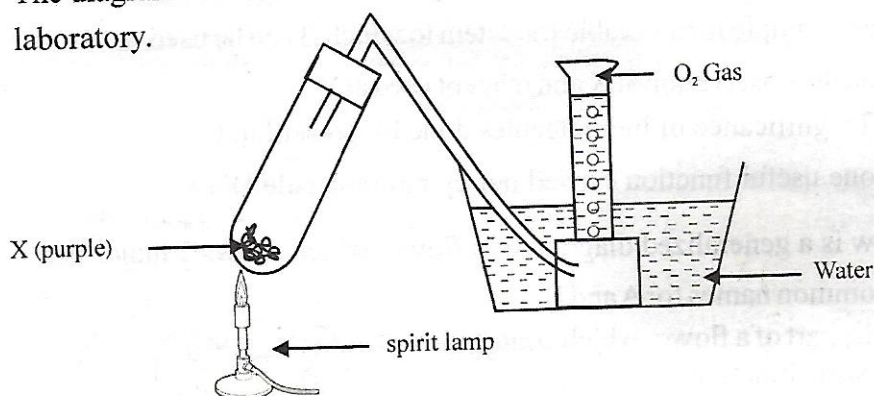


(06) (A) Given below are 4 experimental setups prepared by students for an activity.



- (i) Out of A, B, C, D setups
 - (a) Which setup produce the white color precipitate? (M. 01)
 - (b) What is the setup that produce a gaseous product? (M. 01)
- (ii) What is the colour change that occur in the CuSO_4 solution of setup B? (M. 01)
- (iii) Mention the method of increasing the rate of reaction setup D (M. 01)
- (iv) Write an observation which occurs at A (M. 01)
- (v) Mention how a chemical change differ from a physical change (M. 01)

(B) The diagram below shows a setup made to collect a sample of oxygen gas in laboratory.



- (i) Write the chemical formula of the substance 'X' (M. 01)
 - (ii) Name the method of collecting gas by this method. (M. 01)
 - (iii) State 2 physical characteristics of oxygen gas. (M. 02)
 - (iv) Write 2 uses of oxygen gas. (M. 02)
- (C) Activity Series is obtained by arranging metals in the descending order of their reactivity
It is given below

K, Na, Ca, Mg, Al, Zn, Fe, Sn, Pb, H, Cu, Ag, Pt, Au

- (i) Write 2 metallic reactions shown by metals when forming the activity series. (M. 02)
- (ii) What is the metal which has a higher reactivity than iron but less corrosive (M. 01)
- (iii) Briefly explain how stripe of magnesium prevents iron from corrosion when they are in contact with each other (M. 01)
- (iv) How does sodium metal store? (M. 01)

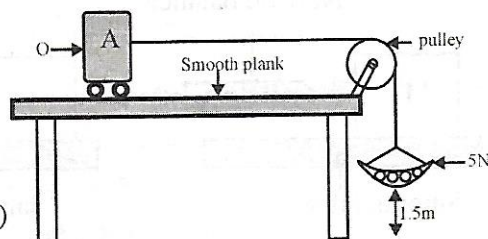
(D) The table shows the information of extracting 3 types of metals

Metal	Method of extraction	Raw materials
Na		NaCl
Fe (Iron)	reduction	Heamatite/ coke / limestone
Au	Physical methods	gold mixed soil

- Name a physical method used to extract Au. (M. 01)
- Al though the temperature of hot air allowed into the blast furnace is 600°C - 800°C temperature rises up to 1400°C within it. State the reason . (M. 01)
- What is the method of a Sodium extraction (M. 01)

(Total marks - 20)

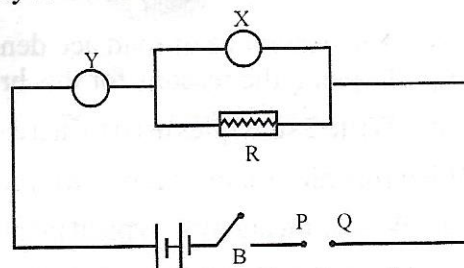
(07) (A) Trolley A is kept on a smooth wooden plank and connected to a pulley by a string as shown. When the trolley is released it started to move slowly.



- What is the limiting frictional force applied on the wheels of the trolley? (M. 01)
- What can be done to speed up the motion of the trolley? (M. 01)
- Copy the trolley in your answer script and mark two forces acting on it being held by hand at '0' (at rest) (M. 02)
- Calculate the potental energy of the 5N load if it is found 1.5m above the ground when the trolley is at rest ($g = 10\text{ms}^{-2}$) (M. 02)

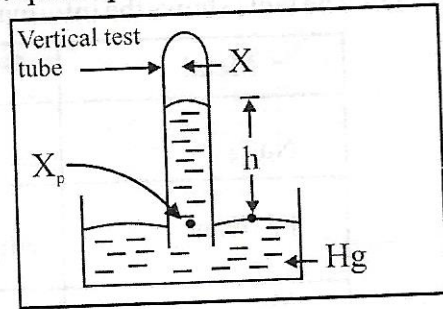
(B) Following is a diagram of a circuit used in an activity to find the value of resist or 'R'

- Name the equipment X and Y (M. 02)
- State a factor that should be considered when connecting X and Y equipment in to the circuit (M. 01)



- The teacher has instructed to disconnect the switch B immediately after taking a reading. What is the reason for this? (M. 01)
- What is the other equipment that should be connected between P and Q in addition to Y and X to get readings at several instances. (M. 01)
- The reading of Y was 0.2 A and X was 4 V at a given instance. Calculate the resistance of 'R' (M. 02)

(C) Pressure is used in our day to day activities. The atmospheric pressure is measured by using a mercury barometer. It can be made by using a glass tube, of 1 m length, with one closed end, mercury and a beaker



(i) Calculate the height of 'h' if atmospheric pressure is $1 \times 10^5 \text{ Pa}$, (Density of mercury is $\rho = 13600 \text{ kg m}^{-3}$, $g = 10 \text{ ms}^{-2}$) (M. 02)

(ii) State an assumption you make here for your calculations. (M. 01)

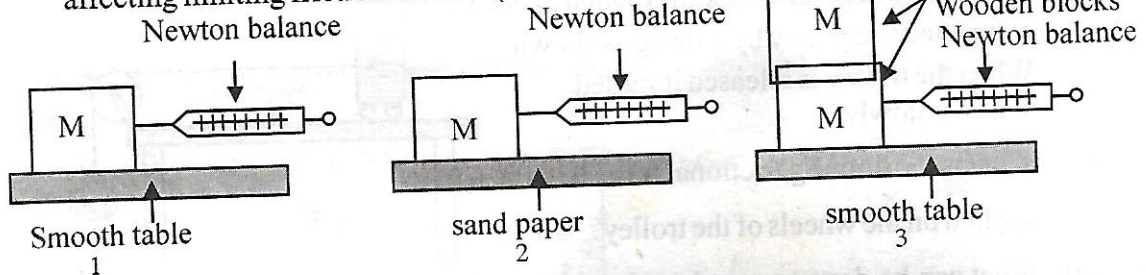
(iii) What can you observe if you take this instrument to a mountainous region about 2000 m above the sea level? (M. 01)

(iv) What is the advantage of using a hydraulic jack (M. 01)

(v) Write 2 uses of hydraulic jak. (M. 02)

(Total marks - 20)

(08) (A) A group of students carried out the following activity to investigate the factors affecting limiting frictional force (M wooden blocks have equal masses)



(i) Name the factors affecting limiting frictional force in the following instances wooden blocks of equal mass are denoted by M

(a) 1st and 2nd instances (M. 02)

(b) 1st and 3rd instances

(ii) Write an expression for the relationship between the mass (m) of the object and its acceleration considering 1 and 2 instances (M. 01)

(iii) Write one instance in which frictional force is advantageous in day to day activities (M. 01)

(iv) The occurrence of road accidents are more in steep roads on rainy days. Explain, the reasons for this briefly (M. 02)

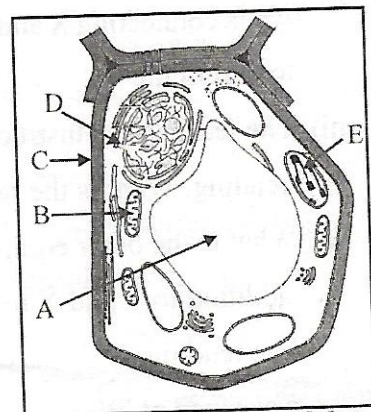
(v) Write 2 strategies used to increase the friction. (M. 02)

(B) Electron microscopic view of a typical cell is given below.

(i) What is meant by a "typical plant cell"? (M. 01)

(ii) Complete the table using the letters indicated in the diagram

Organelle	Function
.....	produce in energy
.....	Maintaining the shape of the cell
.....	Maintaining the water balance



(M. 03)